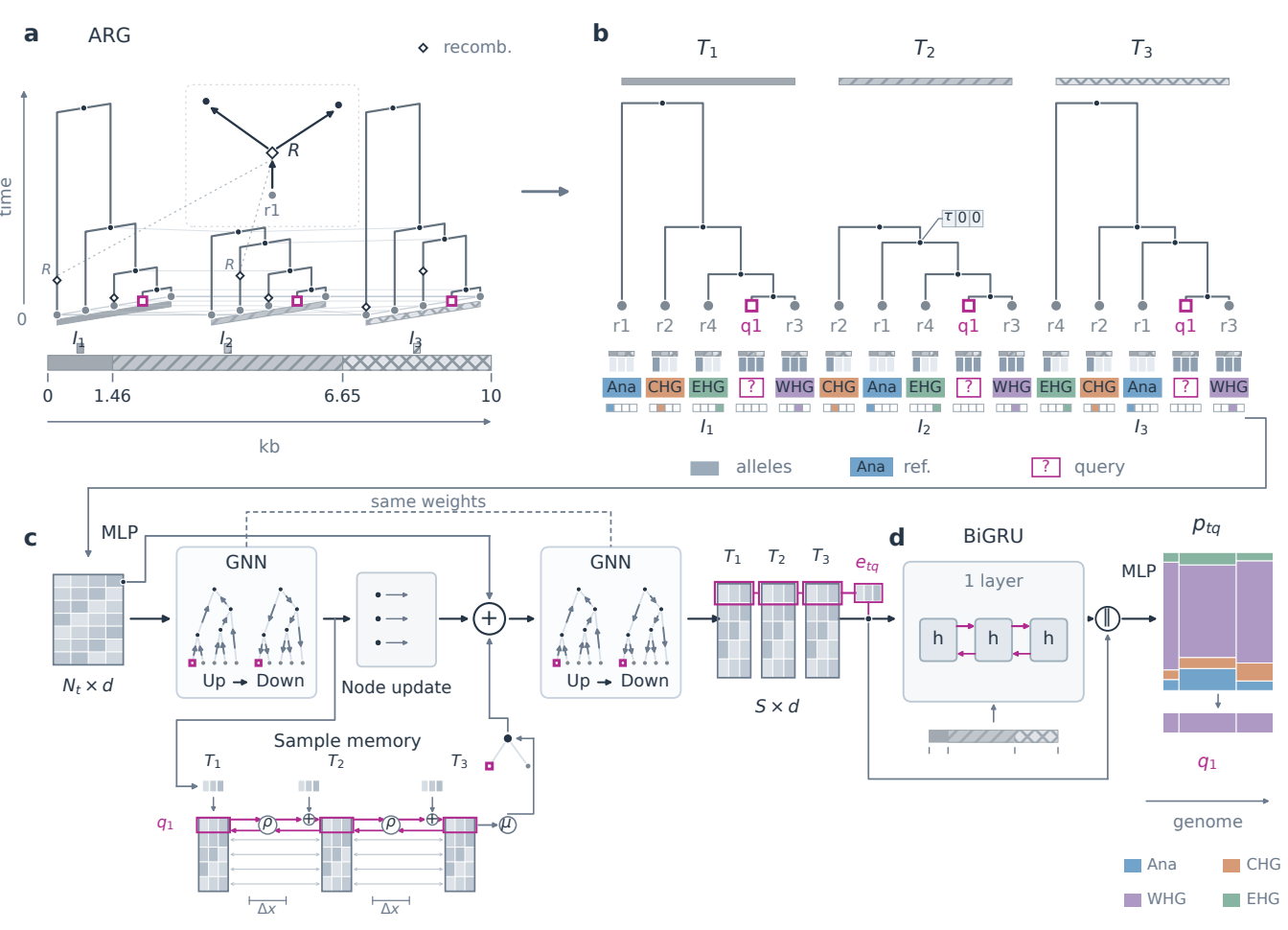

a
ARG
recomb.
time
0
R
r1
l_1
l_2
l_3
0
1.46
6.65
10
kb
b
T_1
T_2
T_3
τ 0 0
r1 r2 r4 q1 r3
Ana CHG EHG ? WHG
alleles
Ana ref.
? query
same weights
c
MLP
$N_t \times d$
GNN
Up → Down
Node update
Sample memory
q_1
Δx
d
BiGRU
1 layer
h
e_{tq}
$S \times d$
MLP
p_{tq}
genome
Ana
CHG
WHG
EHG